Bronze Clinical Genomics

Methods may vary by region. The client portal holds the full, current description of the methods used for this report. Our accreditations and certifications are listed at www.bronze.example.org.

Neurodevelopmental Panel

REFERRING HEALTHCARE PROFESSIONAL

NAME	HOSPITAL
Leah Brennan, MS, CGC	Westbrook Cancer Clinic

PATIENT

NAME	DOB	AGE	GENDER	ORDER ID
Malik Brandt	Oct 12, 2013	12	Male	2399679

PRIMARY SAMPLE TYPE	SAMPLE COLLECTION DATE	CUSTOMER SAMPLE ID
Cheek swab	Feb 05, 2026	

SUMMARY OF RESULTS

PRIMARY FINDINGS

The patient is heterozygous for CHD8 c.3790_3796del, p.(Glu1264fs), which is pathogenic.

PRIMARY FINDINGS: SEQUENCE ALTERATIONS

GENE	TRANSCRIPT	NOMENCLATURE	GENOTYPE	CONSEQUENCE	INHERITANCE	CLASSIFICATION
CHD8	NM_001170629.2	c.3790_3796del, p.(Glu1264fs)	HET	frameshift_variant	AD	Pathogenic

ID	ASSEMBLY	POS
N/A	GRCh38	14:21402422

gnomAD AC/AN	POLYPHEN	SIFT	MUTTASTER	PHENOTYPE
N/A	N/A	N/A	N/A	intellectual developmental disorder with autism and macrocephaly

SEQUENCING PERFORMANCE METRICS

PANEL	GENES	EXONS / REGIONS	BASES	BASES > 20X	MEDIAN COVERAGE	PERCENT > 20X
Neurodevelopmental Panel	10	110	30351	30090	169	99.29

TARGET REGION AND GENE LIST

The genes that the Bronze Clinical Genomics Neurodevelopmental Panel sequences and checks for copy number changes are listed in the client portal at www.bronze.example.org.

STATEMENT

CLINICAL HISTORY

Reason for referral: Global developmental delay.

Patient is a 12-year-old male. Reported features: Pica; Delayed developmental milestones.

CLINICAL REPORT

***CHD8* c.3790_3796del, p.(Glu1264fs)**

The CHD8 variant c.3790_3796del is a deletion of seven bases. Because seven is not a multiple of three, the change shifts the reading frame of the gene. A frameshift like this usually leads to a shortened protein or to a lower amount of the protein. Changes of this kind that stop CHD8 from working are reported in people with this condition. The laboratory classified the CHD8 variant c.3790_3796del as pathogenic.

Other variants in *CHD8*, such as c.6002del, have been reported in unrelated individuals with intellectual developmental disorder with autism and macrocephaly in the medical literature. That variant was not found in this patient.

CONCLUSION

A pathogenic or likely pathogenic variant explains the patient's features. The laboratory classified the CHD8 variant as pathogenic.

The test found one variant, the CHD8 variant c.3790_3796del. It is present in one copy of the gene. The laboratory classified the CHD8 variant c.3790_3796del as pathogenic and concluded that it explains the patient's features.

Changes in CHD8 cause intellectual developmental disorder with autism and macrocephaly. The condition is autosomal dominant, so one changed copy of the gene is enough to cause it. The CHD8 variant c.3790_3796del is consistent with this inheritance pattern, as reported by the laboratory.

Testing of the parents can show whether the CHD8 variant c.3790_3796del was inherited or arose new in the patient. The answer affects the risk for relatives and for future pregnancies.

Test the parents for this CHD8 variant, so that the family can learn whether it was inherited or arose new.

Refer the patient to a clinical geneticist and a neurodevelopmental team to plan care and follow-up of the reported features.

Offer genetic counseling to the family, so that relatives can understand the risk for their own children.

STEP	DATE
Order date	Feb 03, 2026
Sample received	Feb 09, 2026
Reported	Apr 09, 2026

(Our geneticists and physicians reviewed the sequencing results together and wrote this statement.)

Ilse Vartiainen

Ilse Vartiainen, MD, PhD
Laboratory Director

APPENDIX 5: SUMMARY OF THE TEST

Laboratory process. The coding exons and 20 base pairs of flanking intronic sequence of the genes of the panel are captured and sequenced, together with selected non-coding variants that are known to cause disease. Sequencing is performed on a massively parallel sequencing platform.

Bioinformatics and quality control. Reads are aligned to the GRCh38 reference sequence. A validated pipeline calls single nucleotide variants and small indels, and read depth is used to detect copy number variants. Any variant below the quality thresholds is confirmed by Sanger sequencing.

Interpretation. Variants are classified with the ACMG/AMP guidelines into five classes. Benign and likely benign variants are not reported. The classification relies on population data, the literature, computational predictions and the clinical information provided.

Limitations. The test does not detect low-level mosaicism, balanced rearrangements, repeat expansions, or variants in regions of the genome that are not covered with sufficient quality. A negative result does not exclude a genetic cause of the patient's condition.

GLOSSARY

HET	heterozygous
HOM	homozygous
HEM	hemizygous
AD	autosomal dominant
AR	autosomal recessive
XL	X-linked
VUS	variant of uncertain significance
CNV	copy number variation